

Harshit Shah

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PROFILE SUMMARY

AI Engineer with expertise in developing and deploying AI systems leveraging LLMs, ML, and data analysis to drive business value. Proven experience in fine-tuning, prompt engineering, and MLOps to solve real-world problems. Published researcher with a track record of measurable impact across multiple products.

EDUCATION

MIT ADT University, Pune, India

Aug 2021 - Jul 2025

B.Tech in Biomedical Engineering

Relevant Coursework: Python, DBMS, Java, NLP, Data Structures & Algorithms, Linear Algebra, Machine Learning, Artificial Intelligence

SKILLS

Technical : LLM Fine-tuning, Machine Learning, RAG, NL2SQL, Prompt Engineering, MLOps, Agentic AI

AI / ML : PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, LangGraph, OpenCV, Whisper, FastAPI Languages / Tools : Python, SQL, Java, Pinecone, Vector Databases, Docker, AWS, Git/GitHub, CrewAI

PROFESSIONAL EXPERIENCE

Actorius Innovations and Research - Pune AI

Oct 2025 - Present

Engineer

- Engineered a production-deployed WhatsApp AI assistant using LLM + RAG, supporting multi-turn FAQ resolution, automated test booking, and intelligent live-agent escalation for real users at production scale.
- Built a desktop CTC analysis application that processes .czi (Carl Zeiss) microscopy files through a marker-first pipeline, automating fluorescence channel parsing, cell segmentation, and multi-class classification.
- Designed an NL2SQL system using a fine-tuned transformer with schema-aware prompting, enabling the clinical team to query structured patient datasets in plain English without any SQL knowledge.

1Cell.AI - Pune

Jul 2025 - Oct 2025 Machine

Learning Engineer

- Built a document intelligence pipeline processing 500+ oncology PDFs/month using PyPDF2 and pdfplumber with 12+ regex patterns for genomic entity extraction (SNVs, CNVs, fusions, VAF, MSI, TMB, HRD), achieving 95% accuracy and 3x throughput via parallel processing.

1Cell.AI - Pune

Jan 2025 - Jun 2025

Project Intern

- Automated CTC image analysis using ImageJ macros, Java plugins, and Bio-Formats integration, achieving a 140x performance improvement from 7 minutes to 3 seconds per sample; published findings in ESMO Open (2025).

Bioinformatics Centre, SPPU - Pune

Jul 2024 - Sep 2024

Research Intern

- Performed DESeq2 differential expression analysis on 522 TCGA RNA-seq samples, identified 117 DEGs, and surfaced 10 hub gene biomarker candidates via Cytoscape PPI network analysis, validated across 9,736 samples on GEPIA.

PROJECTS

Genetics-GPT – AI-Powered Genomics Question Answering System

- Designed and deployed an AI-driven QA system for genomics using LLaMA 3.1, integrating NCBI and Ensembl APIs to provide real-time gene and research data retrieval.
- Implemented gene mention detection, biomedical entity linking, and dynamic summarization for high-accuracy answers based on scientific literature.
- Integrated voice-based interaction features to support hands-free usage in laboratory settings, enhancing accessibility and productivity for researchers

PUBLICATIONS

Bhosale, B., Shah, H., et al. (2025). *Profiling of PD-L1 and HER2 overexpression on cancer cells using AI-based macro-driven automation*. ESMO Open, 10(3), 105588. DOI: [10.1016/j.esmoop.2025.105588](https://doi.org/10.1016/j.esmoop.2025.105588)

ACHIEVEMENTS

- Finalist, NABARD Rural Innovation Hackathon (2024).